

HUMAN HAND GESTURES BASED AND VOICE CONTROLLED VIDEO PLAYER

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ABSTRACT

In this paper we have discussed a low cost system which uses openCV and NPL to control the VLC media player through human based hand gesture and speech. Human hand gesture recognition (HGR) is one of the main topics in the area of computer science as an effective approach for HCI(Human Computer Interaction). Our proposed project is an alternative to this, wherein we are developing a new technique to interact with the computer in a non-tangible way. This project describes about controlling video using gestures and speech. Instead of using a keyboard, mouse or joystick, we can use our hand gestures to control certain functions of a computer like to play/pause a video, increasing and decreasing of volume and forward and backward of video. This project also helps in controlling the VLC player using speech. This application uses the webcam to capture gesture by the user also uses microphone to recognize the speech and performs basic operations such as play and pause, volume controls ,forward and backward of the video. This helps the easy accessing the playback without hardware .This even helps the people with disability people for hand fractured or leg fractured to control video through Speech.

1. INTRODUCTION

With the massive influx and advancement of technologies, a computer system has become a very powerful machine which has been designed to make the human beings' tasks easier. Due to which the HCI (human – computer interaction) has become an important part of our lives.

Now-a-days, the progress and development in interaction with computing devices has increased so fast that as a human being even we could not remained left with the effect of this and it has become our primary thing. The technologies has so much surrounded us and has made a place in our lives that we use it to communicate, shop, work and even entertain ourselves¹. There are many applications like media player, MS-office, Windows picture manager etc. which require natural and intuitive interface. Now-a-days most of the users uses keyboard, mouse, pen, Joysticks etc. to interact with computers, which are not enough for them. In the near future, these existing technologies which are available for the computing, communication and display will become a bottleneck and the advancement in these technologies will be required to make the system as natural as possible. Nevertheless the invention of mouse and keyboards by the researchers and engineers has been a great progress, there are still some situations where interaction with computer with the help of keyboard and mouse will not be enough.

As computer innovation keeps on developing, the requirement for characteristic correspondence amongst people and machines additionally increments. In spite of the fact that our cell phones influence utilization of the touch to screen innovation, it is not sufficiently shabby to be actualized in work area frameworks. In spite of the fact that the mouse is exceptionally valuable for gadget control, it could be badly arranged

to use for physically disabled individuals and individuals who are not familiar with utilizing the mouse for connection. The strategy proposed in this paper makes utilization of a webcam through which hand gestures given by the user are captured and identified accordingly. Hand gestures have boundless applications. In this investigation, we apply it to a system to make a straightforward easy to understand interaction interface. The research paper sheds light on the recognition of letters from the hand language which is taken as per the American Sign Language. The detection is done using the various techniques of Contour Analysis and Feature Extraction. The research paper sheds light on the recognition of letters from the hand language which is taken as per the American Sign Language. The detection is done using the various techniques of Contour Analysis and Feature Extraction

Gesture recognition is a topic in computer science and language technology with the goal of interpreting human gestures via mathematical algorithms. Gestures can originate from any bodily motion or state but commonly originate from the face or hand. Current focuses in the field include emotion recognition from face and hand gesture recognition. Users can use simple gestures to control or interact with devices without physically touching them. Many approaches have been made using cameras and computer vision algorithms to interpret sign language. However, the identification and recognition of posture, gait, proxemics, and human behaviors is also the subject of gesture recognition techniques. Gesture recognition can be seen as a way for computers to begin to understand human body language, thus building a richer bridge between machines and humans than primitive text user interfaces or even GUIs (graphical user interfaces), which still limit the majority of input to keyboard and mouse and interact naturally without any mechanical devices. Using the concept of gesture recognition, it is possible to point a finger at this point will move accordingly. This could make conventional input on devices such and even redundant.

Gesture recognition features:

- More accurate
- High stability
- Time saving to unlock a device

Gesture types

In computer interfaces, two types of gestures are distinguished. We consider online gestures, which can also be regarded as direct manipulations like scaling and rotating. In contrast, offline gestures are usually processed after the interaction is finished; e. g. a circle is drawn to activate a context menu.

- **Offline gestures:** Those gestures that are processed after the user interaction with the object. An example is the gesture to activate a menu.
- **Online gestures:** Direct manipulation gestures. They are used to scale or rotate a tangible object.

Touchless Interface

Touchless user interface is an emerging type of technology in relation to gesture control. Touchless user interface (TUI) is the process of commanding the computer via body motion and gestures without touching a keyboard, mouse, or screen. Touchless interface in addition to gesture controls are becoming widely popular as they provide the abilities to interact with devices without physically touching them.

Types of Touchless Technology

There are a number of devices utilizing this type of interface such as, smartphones, laptops, games, television, and music equipment.

Input Devices

The ability to track a person's movements and determine what gestures they may be performing can be achieved through various tools. The kinetic user interfaces (KUIs) are an emerging type of user interfaces that allow users to interact with computing devices through the motion of objects and bodies. Examples of KUIs include tangible user interfaces and motion-aware games such as Wii and Microsoft's Kinect, and other interactive projects.

Although there is a large amount of research done in image/video based gesture recognition, there is some variation within the tools and environments used between implementations.

- **Wired gloves** These can provide input to the computer about the position and rotation of the hands using magnetic or inertial tracking devices. Furthermore, some gloves can detect finger bending with a high degree of accuracy (5-10 degrees), or even provide haptic feedback to the user, which is a simulation of the sense of touch. The first commercially available hand-tracking glove-type device was the DataGlove, a glove-type device which could detect hand position, movement and finger bending. This uses fiber optic cables running down the back of the hand. Light pulses are created and when the fingers are bent, light leaks through small cracks and the loss is registered, giving an approximation of the hand pose.

- **Depth-aware cameras** Using specialized cameras such as structured light or time-of-flight cameras, one can generate a depth map of what is being seen through the camera at a short range, and use this data to approximate a 3d representation of what is being seen. These can be effective for detection of hand gestures due to their short range capabilities.

- **Stereo cameras** Using two cameras whose relations to one another are known, a 3d representation can be approximated by the output of the cameras. To get the cameras' relations, one can use a positioning reference such as a lexian stripe or infrared emitters. In combination with direct motion measurement (6D Vision) gestures can directly be detected.

- **Gesture-based controllers** These controllers act as an extension of the body so that when gestures are performed, some of their motion can be conveniently captured by software. An example of emerging gesture-based motion capture is through skeletal hand tracking, which is being developed for virtual reality and augmented reality applications. An example of this technology is shown by tracking companies uSens and Gestigon, which allow users to interact with their surrounding without controllers. Another example of this is mouse gesture trackings, where the motion of the mouse is correlated to a symbol being drawn by a person's hand which can study changes in acceleration over time to represent gestures. The software also compensates for human tremor and inadvertent movement. The sensors of these smart light emitting cubes can be used to sense hands and fingers as well as other objects nearby, and can be used to process data. Most applications are in music and sound synthesis, but can be applied to other fields.

- **Single camera** A standard 2D camera can be used for gesture recognition where the resources/environment would not be convenient for other forms of image-based recognition. Earlier it was thought that single camera may not be as effective as stereo or depth aware cameras, but some companies are challenging this theory. Software-based gesture recognition technology using a standard 2D camera that can detect robust hand gestures.

Speech recognition is an interdisciplinary subfield of computer science and computational linguistics that develops methodologies and technologies that enable the recognition and translation of spoken language into text by computers. It is also known as automatic speech recognition (ASR), computer speech recognition or speech to text (STT). It incorporates knowledge and research in the computer science, linguistics and computer engineering fields.

Some speech recognition systems require "training" (also called "enrollment") where an individual speaker reads text or isolated vocabulary into the system. The system analyzes the person's specific voice and uses it to fine-tune the recognition of that person's speech, resulting in increased accuracy. Systems that do not use training are called "speaker independent" systems. Systems that use training are called "speaker dependent".

Speech recognition applications include voice user interfaces such as voice dialing (e.g. "call home"), call routing (e.g. "I would like to make a collect call"), domestic appliance control, search key words (e.g. find a podcast where particular words were spoken), simple data entry (e.g., entering a credit card number), preparation of structured documents (e.g. a radiology report), determining speaker characteristics, speech-to-text processing (e.g., word processors or emails), and aircraft (usually termed direct voice input).

The term Voice recognition or speech identification refers to identifying the speaker, rather than what they are saying. Recognizing the speaker can simplify the task of translating speech in systems that have been trained on a specific person's voice or it can be used to authenticate or verify the identity of a speaker as part of a security process.

From the technology perspective, speech recognition has a long history with several waves of major innovations. Most recently, the field has benefited from advances in deep learning and big data. The advances are evidenced not only by the surge of academic papers published in the field, but more importantly by the worldwide industry adoption of a variety of deep learning methods in designing and deploying speech recognition systems.

The paper invokes the use of various computer vision techniques and algorithms which are involved in the determination of hand gestures.

2. BACKGROUND AND RELATED WORK

Facing these limitations of traditional Remote controls are commonly used today as an interface for almost all consumer electronic equipment. Different home appliances use a multitude of remote controllers, offering a variety of problems: confusion in the different button layout, spending time in looking for the right remote, problems in replacement when destroyed and so on.

Present-day remote controls are commonly consumer infrared devices which send digitally-coded pulses of infrared radiation to control functions such as power, volume, channels, playback, track change, heat, fan speed, or other features varying from

device to device. Remote controls for these devices are usually small wireless handheld objects with an array of buttons for adjusting various settings such as television channel, track number, and volume. For many devices, the remote control contains all the function controls while the controlled device itself has only a handful of essential primary controls. The remote control code, and thus the required remote control device, is usually specific to a product line, but there are universal remotes, which emulate the remote control made for most major brand devices.

Problems in Existing System

- When objects block the infrared light it causes difficulties.
- Need batteries
- Electricity
- Small buttons
- Disable people to operate devices
- Hardware
- Consumes more time

With the development of today's technology, and as humans tend to naturally use hand gestures in their communication process to clarify their intentions, hand gesture recognition is considered to be an important part of Human Computer Interaction (HCI), which gives computers the ability of capturing and interpreting hand gestures, and executing commands afterwards. The aim of this study is to perform a systematic literature review for identifying the most prominent techniques, applications and challenges in hand gesture recognition. To conduct this systematic review, we have screened 560 papers retrieved from IEEE Explore published from the year 2016 to 2018, in the searching process keywords such as "hand gesture recognition" and "hand gesture techniques" have been used. However, to focus the scope of the study 465 papers have been excluded. Only the most relevant hand gesture recognition works to the research questions, and the well-organized papers have been studied. The results of this paper can be summarized as the following; the surface electromyography (sEMG) sensors with wearable hand gesture devices were the most acquisition tool used in the work studied, also Artificial Neural Network (ANN) was the most applied classifier, the most popular application was using hand gestures for sign language, the dominant environmental surrounding factor that affected the accuracy was the background color, and finally the problem of overfitting in the datasets was highly experienced. The paper will discuss the gesture acquisition methods, the feature extraction process, the classification of hand gestures, the applications that were recently proposed, the challenges that face researchers in the hand gesture recognition process, and the future of hand gesture recognition. We shall also introduce the most recent research from the year 2016 to the year 2018 in the field of hand gesture recognition for the first time.

Machine Learning of Musical Gestures

We present an overview of machine learning (ML) techniques and their application in interactive music and new digital instrument design. We first provide the non-specialist reader an introduction to two ML tasks, classification and regression, that are particularly relevant for gestural interaction. We then present a review of the literature in current NIME research that uses ML in musical gesture analysis and

gestural sound control. We describe the ways in which machine learning is useful for creating expressive musical interaction, and in turn why live music performance presents a pertinent and challenging use case for machine learning.

Hand Gesture Detection and Conversion to Speech and Text

The hand gestures are one of the typical methods used in sign language. It is very difficult for the hearing impaired people to communicate with the world. This project presents a solution that will not only automatically recognize the hand gestures but will also convert it into speech and text output so that impaired person can easily communicate with normal people. A camera attached to computer will capture images of hand and the contour feature extraction is used to recognize the hand gestures of the person. Based on the recognized gestures, the recorded soundtrack will be played.

Survey on Emotional Body Gesture Recognition

Automatic emotion recognition has become a trending research topic in the past decade. While works based on facial expressions or speech abound recognizing affect from body gestures remains a less explored topic. We present a new comprehensive survey hoping to boost research in the field. We first introduce emotional body gestures as a component of what is commonly known as "body language" and comment general aspects as gender differences and culture dependence. We then define a complete framework for automatic emotional body gesture recognition. We introduce person detection and comment static and dynamic body pose estimation methods both in RGB and 3D. We then comment the recent literature related to representation learning and emotion recognition from images of emotionally expressive gestures. We also discuss multi-modal approaches that combine speech or face with body gestures for improved emotion recognition. While pre-processing methodologies (e.g. human detection and pose estimation) are nowadays mature technologies fully developed for robust large scale analysis, we show that for emotion recognition the quantity of labelled data is scarce, there is no agreement on clearly defined output spaces and the representations are shallow and largely based on naive geometrical representations.

Gesture and speech in interaction

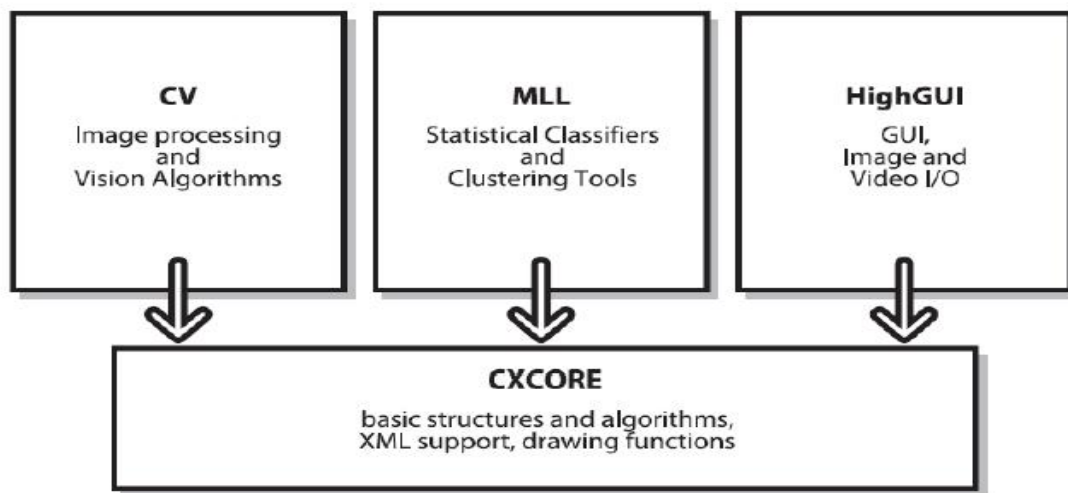
Gestures and speech interact. They are linked in language production and perception, with their interaction contributing to felicitous communication. The multifaceted nature of these interactions has attracted considerable attention from the speech and gesture community. This article provides an overview of our current understanding of manual and head gesture form and function, of the principle functional interactions between gesture and speech aiding communication, transporting meaning and producing speech. Furthermore, we present an overview of research on temporal speech-gesture synchrony, including the special role of prosody in speech-gesture alignment. In addition, we provide a summary of tools and data available for gesture analysis, and describe speech-gesture interaction models and simulations in technical systems. This overview also serves as an introduction to a Special Issue covering a wide range of articles on these topics. We provide links to the Special Issue throughout this paper.

3. METHODS

Software requirements and specification or set of documents that describes the features and behavior of the system or system applications. It includes a variety of elements that attempts to define the intended functionality required by the customer to satisfy their different users.

3.1 Technologies Used

Open CV



OpenCV Structure and Content

OpenCV can be broadly structured into five primary components, four of which are shown in the figure. The CV component contains mainly the basic image processing and higher-level computer vision algorithms; MLL the machine learning library includes many statistical classifiers as well as clustering tools. HighGUI component contains I/O routines with functions for storing, loading video & images, while CXCore contains all the basic data structures and content.

Speech Recognition

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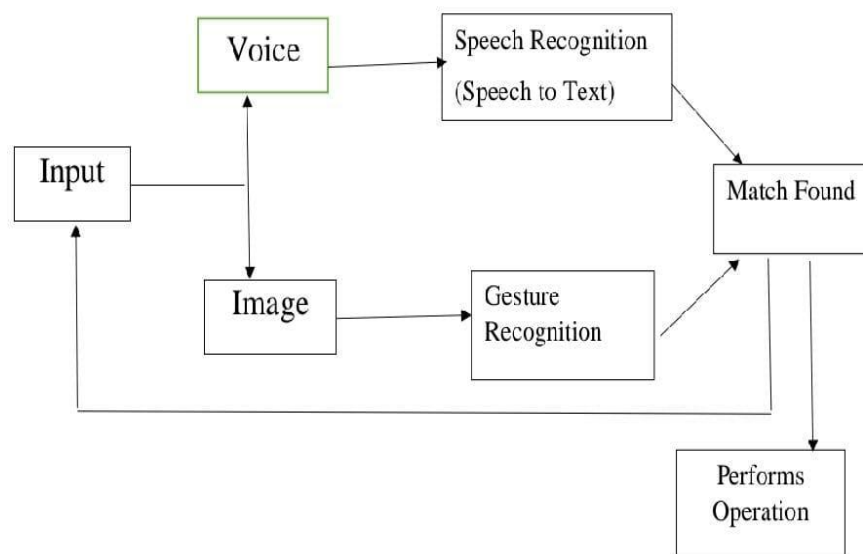
called "speaker independent" systems. Systems that use training are called "speaker dependent".

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4. SYSTEM DESIGN

Architecture of Proposed System

According to architecture, where user gives input to the system by using camera or speaker. After successful identification of user input it perform the respective operation. If a user doesn't give correct input then he/she cannot see the operation performed on computer.



Architecture of Proposed System

For new user or old user , he/she can keep there hands in front of camera to detect their hands,Even we can use the microphone to speak where the user can speak the instruction to operate the video. These details are stored in the database and conforms the match operation and performs the operation.

Modules

It is a separable component, frequently one that is interchangeable with others, for assembly into units of differing size, complexity, or function.

There are four modules are present the following are:

Background Learning Module

This is the first module in the system whose main purpose is to know about the background in a sense that how it is different from the foreground. Furthermore as

proposed system works on a video feed, this module extracts the frames from it and learns about the background.

Foreground Extraction Module

This module consists of three steps, background subtraction, image enhancement and foreground extraction. Background is subtracted so that foreground objects are visible.

Gesture Recognition Module

Gesture recognition systems usually are separated into three components: image pre processing, features extraction, and pattern recognition. Computer vision based systems use data obtained from video cameras as input, which is pre processed, eliminating noise and highlighting interest regions. This data is used as input to a feature extraction algorithm that generates domain-based feature vectors to be employed in the classification module. A classification module receives these features and estimates to which class they belong.

Speech Recognition Module

Library for performing speech recognition, with support for several engines and APIs, online and offline. To use this module, we have to install the SpeechRecognition module. For External Microphones or USB microphones, we need to provide the exact microphone to avoid any difficulties. On Linux, if we type 'lsusb' to show the related information for USB devices. Take different microphone related information. configure the microphone using chunk size, sampling rate, ambient noise adjustments etc. Wait for some time to get the voice. When the voice is recognized, try to convert it into texts, otherwise raise some errors. Stop the process.

5. IMPLEMENTATION

gesture.py:

- Uses OpenCV and Handy library for hand gesture recognition.
- Displays instructions for gestures (volume control, playback).
- Initializes a VLC media player for video playback.
- Captures hand histogram for improved hand detection.
- Utilizes a loop for real-time gesture recognition and corresponding actions (volume control, seek, etc.).

handy.py:

- Defines a Hand class with methods for drawing outlines and obtaining the center of mass of the hand.
- Implements functions for face detection and hand segmentation using computer vision techniques.

Voice.py:

- Integrates speech recognition (SpeechRecognition library) for voice commands.
- Creates a list of available videos and initializes VLC media player.
- Recognizes voice commands such as play, pause, forward, backward, increase/decrease volume, etc.

6. RESULTS

6.1 Gesture Recognition

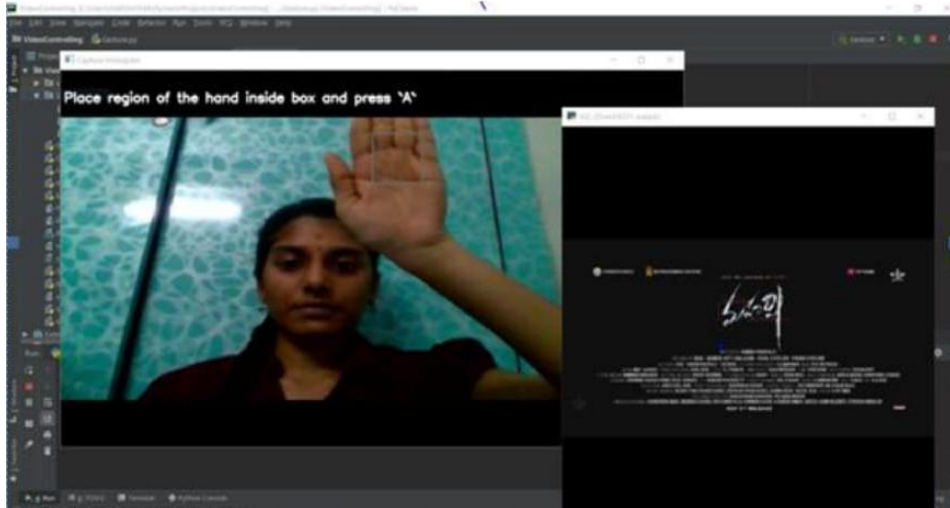


Fig (a) Login

To operate the video, user need to run the program in the pycharm compiler. After the program runs the Capture Histogram dialog box get appear and VLC dialog box also gets appear simultaneously.

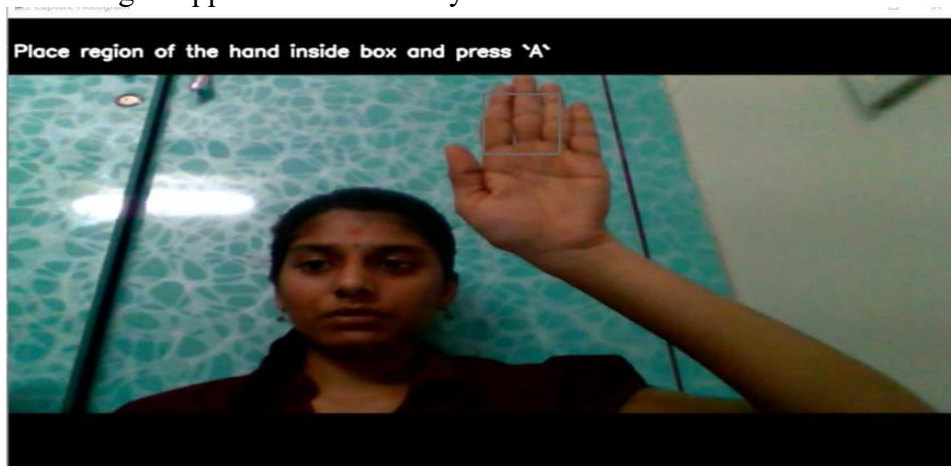


Fig (b) Capture Histogram

In the Capture Histogram ,If new user or any other user need to control then they can place the their hand in the box region and press A where the webcam captures or detects the hand.

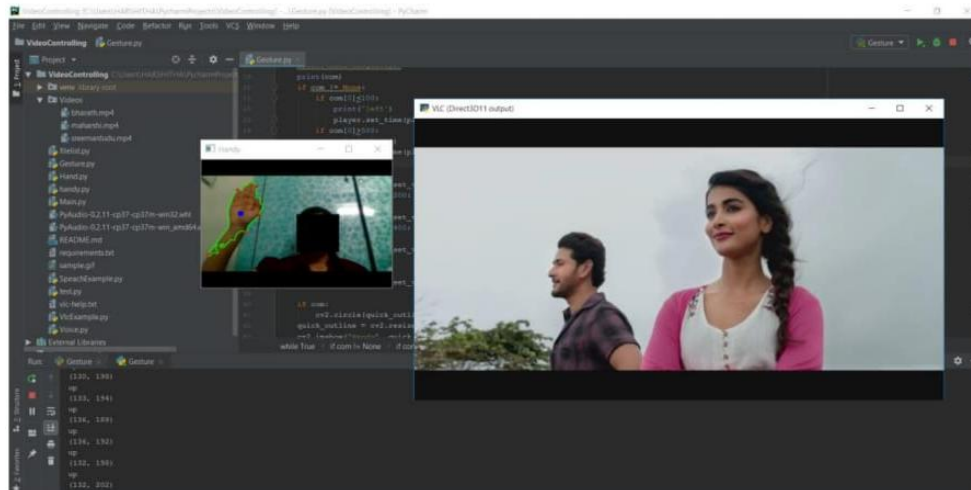


Fig (c) Move upside

To Increase Volume through Gesture, where after the hand gets recognize the hand gets change to blue dot where this resembles the hand and face gets black box .So, that hand and face doesn't get match.the hand moved left side of the dialog box then the video in VLC dialog box moves backward.

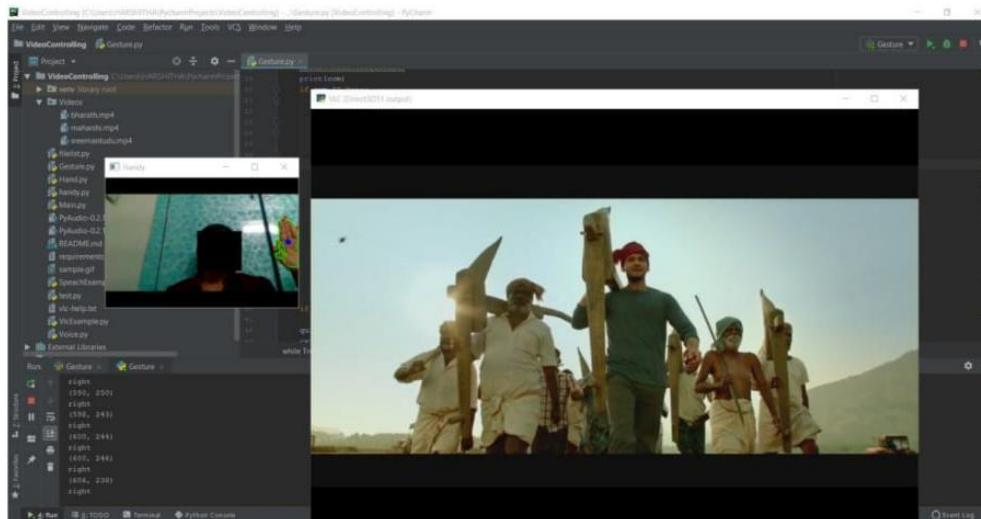


Fig (d) Move Right side

In video controlling to move forward , the user need to move is hand to the rights side so that the video adjust its latitude of x, y axis and does the operation and helps in moving video forward.

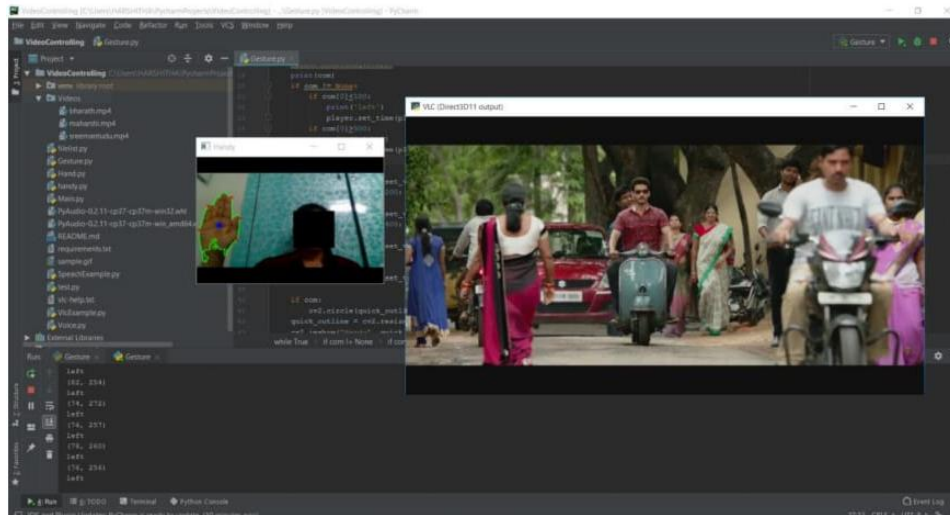


Fig (e) Move Left side

To Move Backward through Gesture , where after the hand gets recognize the hand gets change to blue dot where this resembles the hand and face gets black box .So, that hand and face doesn't get match. The hand moved left side of the dialog box then the video in VLC dialog box moves backward.

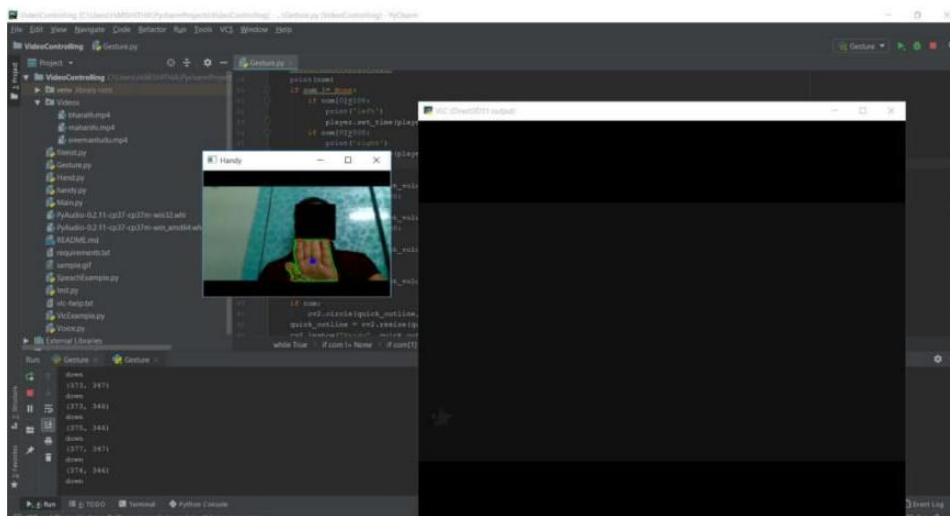


Fig (f) Move Downwards

To Decrease the Volume , the user need to move the hand downwards so that the sound of the video decrease.

6.2 Voice Recognition

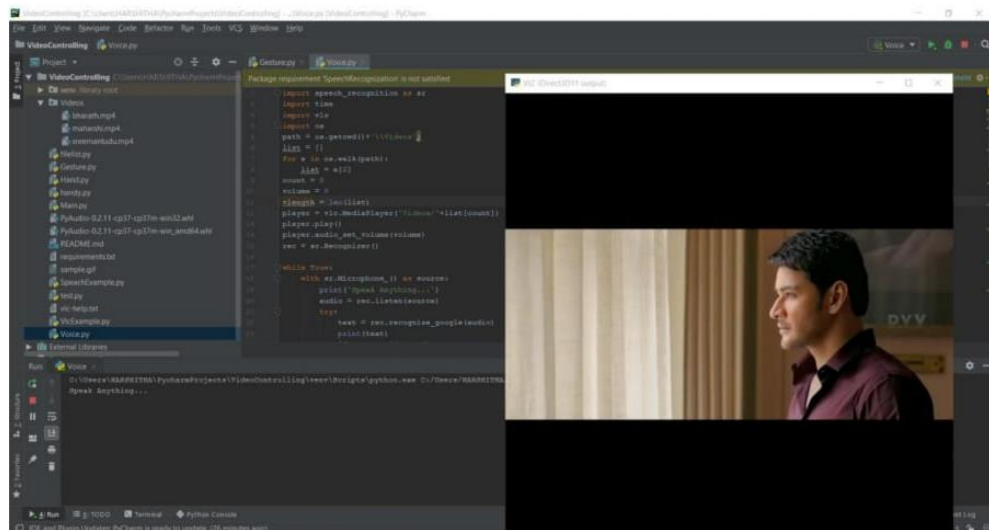


Fig (g) Video controlling through voice

To control the video through voice where we need to open the voice.py file in pycharm then run the code in the console we will be get a statement speak anything where we need to speak the commands to control the video. So the above figure show you how the the terminal of the voice.py program looks after we run it. We get VLC dialog box gets open and terminal runs. In video controlling to pause the video. We need to put Speak the command “pause “ so that the video stops.

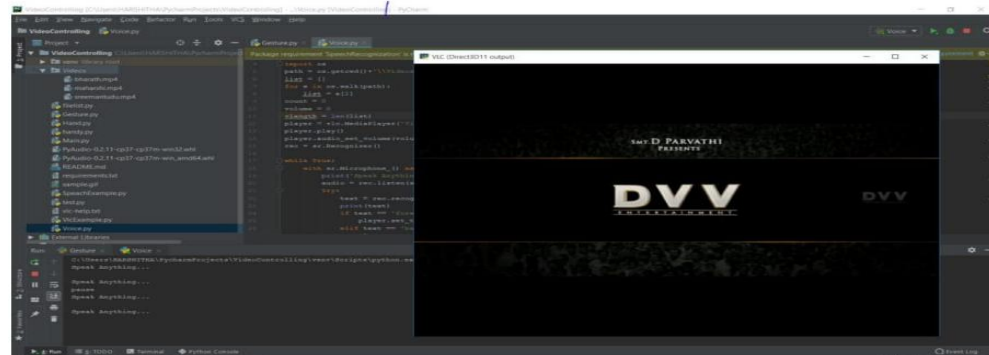


Fig (h) Pause

To play the video through voice then we need to give the another command after it asks again to “speak anything” then if we say to “play”, then video plays.

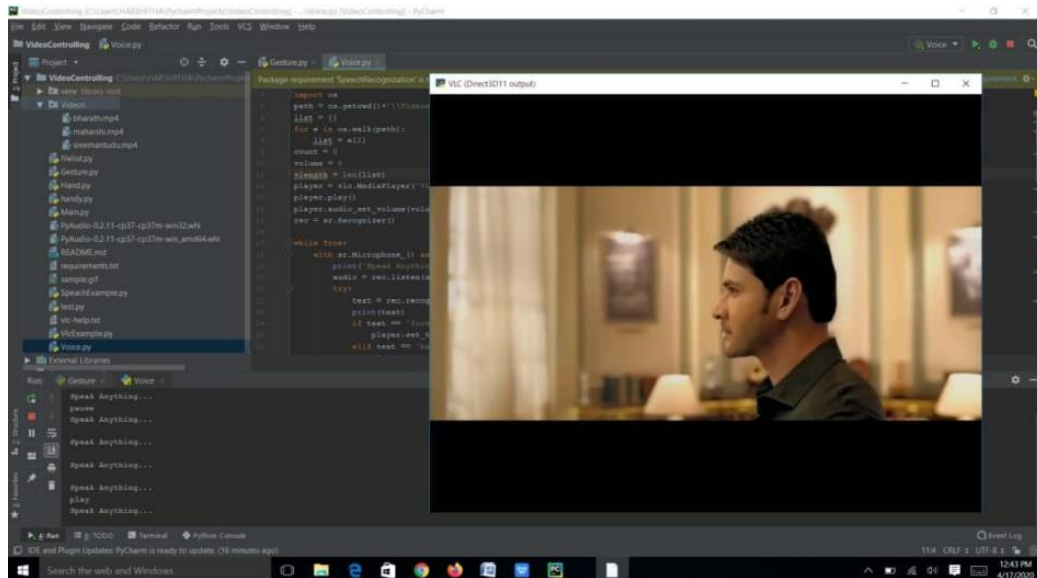


Fig (i) Play

In increase volume we need to just give the other command to the terminal which helps the video to increase the volume by 25% each time of giving the command.

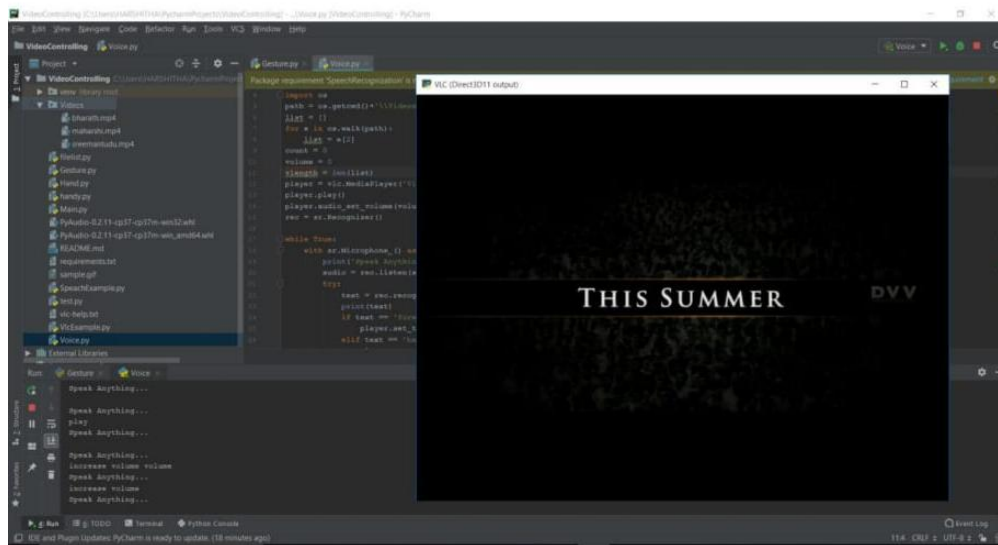


Fig (j) Increase Volume

To decrease the volume through gestures we need to give a command “Decrease Volume” this help he video sound to decrease.

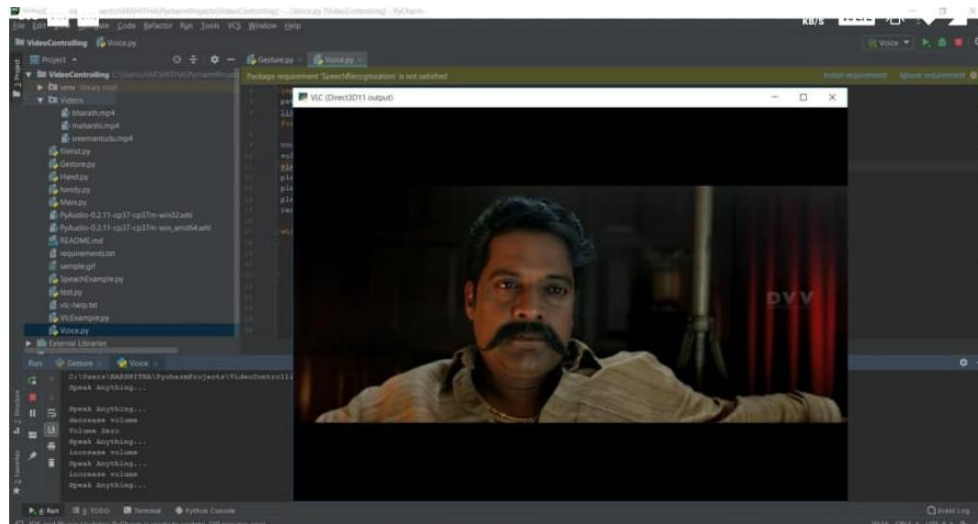


Fig (k) Decrease Volume

To move forward we need to give the command to “move forward ” which helps us to move 10 seconds forward.

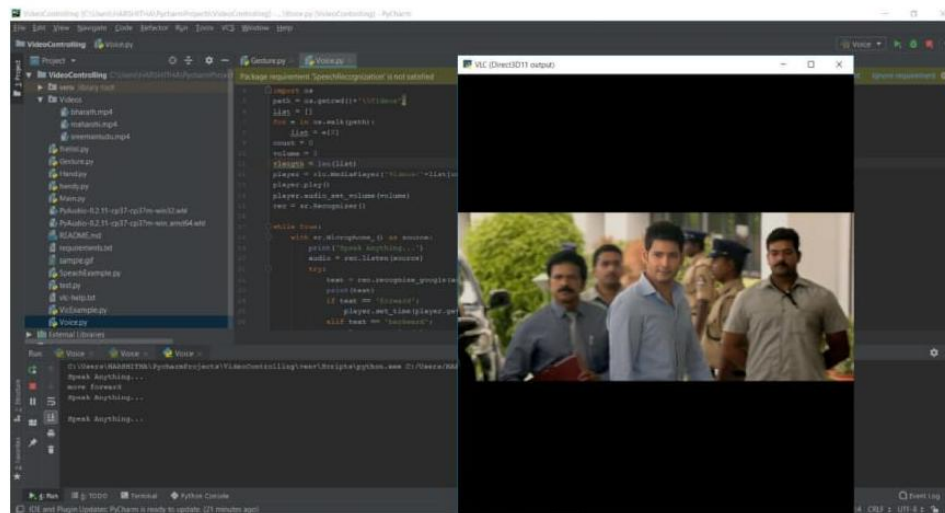


Fig (l) Move forward

To move backward we need to give the command to “move backward” in console where the video moves backward.

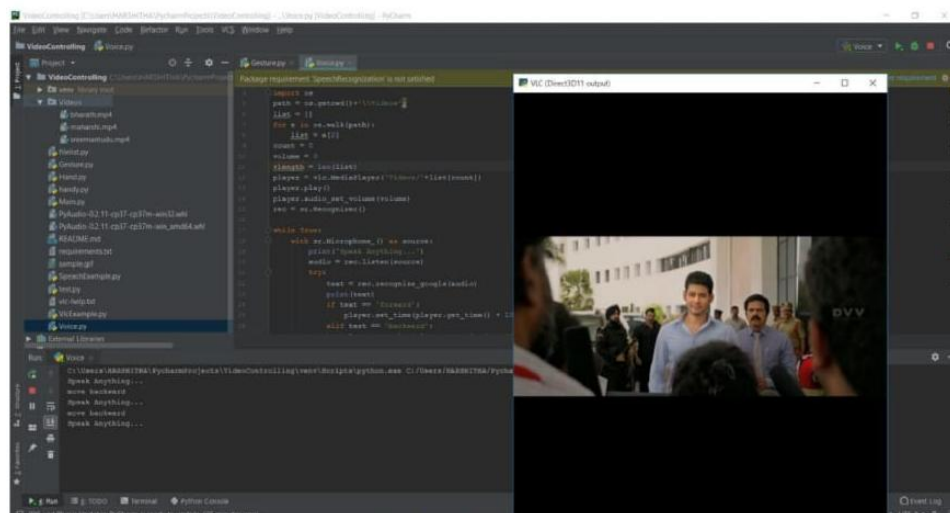


Fig (m) Move backward

7. RESULT ANALYSIS

In the present world, managing diversity is under threat, especially when impaired people are involved.

The present machine usage focuses on its efficiency and its sophisticated design. The recent modules involve usage of equipment viz. gloves, pointers to help in hand movement in applications involving communication. This algorithm can be used by the people who are deaf-mute. It provides a medium for these people to interact with the outside world. It can be used to convert hand gestures both into speech and text. Not only it can be used for the visually impaired, but also by a common man to convert his actions into text or speech based on his requirement.

The proposed solution falls between visually based interfaces. The image of the hand is taken through the camera and is converted into a binary image. The binary image is basically monochromatic and can be processed. Hence, the OpenCV library is optimized and used for such actions.

8.CONCLUSION

In this work, a gesture synthesizer based an audio-visual correlation is presented. Audio-visual correlation analysis is conducted using a camera and micro phone. Visual events are selected as hand gestures. The types of events are defined by investigating a portion of the database. The repetitive patterns for micro phone events are mainly keywords (move forward, move backward , increase volume, decrease volume, pause, play) and accents. The repetitive patterns for hand gestures are moving left ,moving right, and up and down .

The practical adaption of the interface solution for visually impaired and blind people is limited by simplicity and usability in practical scenarios. As an easy and practical way to achieve human-computer- interaction, in this solution hand gesture to speech and text conversion has been used to facilitate the reduction of hardware components. On the whole, the solution aims to provide aid to those in need thus ensuring social relevance. The people can easily communicate with each other. The user-friendly

nature of the system ensure that people can use it without any difficulty and complexity. The application is cost efficient and eliminates the usage of expensive technology.

CONTRIBUTION

In this project, I contributed in implementing the gesture recognition system using OpenCV and the Handy library. The gesture.py file encapsulates the core logic for capturing hand gestures, and I extended it to include functionalities like volume control and playback actions based on recognized gestures. Including computer vision techniques, I contributed to the development of the handy.py module, specifically the Hand class, which facilitated hand attribute handling. Additionally, I focused on writing the introductory sections and the background related work of the report.

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